

Maitree Varia

San Diego, CA • +1 (415) 791-2677 • mvaria@ucsd.edu • [linkedin.com/in/maitree-varia](https://www.linkedin.com/in/maitree-varia) • github.com/MaitreeVaria

EDUCATION

University of California, San Diego

San Diego, CA

- **Master of Science in Computer Science** | Artificial Intelligence | GPA: 3.94 / 4.00 09/2024 - 06/2026
- **Graduate Student Researcher**, JET Lab | **Software Engineer**, KNOLO Agents (UCSD Health-Affiliated Research)
- **Teaching Assistant**, VIS 142: Art + Computing | **Undergrad Mentor**

NMIMS University, Mukesh Patel School of Technology Management & Engineering

Mumbai, India

- **Bachelor of Technology in Computer Engineering** | Machine Learning | GPA: 3.92 / 4.00 07/2020 - 05/2024
- **Dean's List** (*all semesters*)
- **Cohort President**, Student Govt. Association 2021-2023 | **President of People and Culture** 2020
- Research in **Federated Learning** (2 publications, IEEE & AIP)

SKILLS

Programming & Web Technologies: Python, Java, JavaScript, TypeScript, SQL, GraphQL, HTML, CSS, React, Angular, Node.js, Express, Django, MongoDB, PostgreSQL, PHP, Flask, Neo4j

AI & Machine Learning: LangGraph, LanChain, Agentic AI, Retrieval-Augmented Generation, Gen AI, LLM, TensorFlow, Keras, scikit-learn, Computer Vision

Cloud & Distributed Systems: AWS (Lambda, EC2, SNS, S3, DynamoDB, Bedrock Agents, OpenSearch), Redis, REST APIs, Apache Kafka, Google Cloud Platform, Microservices, Serverless Architecture

WORK EXPERIENCE

Aptino, Inc.

Irving, TX

Software Development Engineer Intern

06/2025 - 09/2025

- Architected a **distributed multi-agent system** using **LangGraph + Whisper** for audio interactions in facilities operations, saving managers **3+ hours per week** by eliminating manual coordination of service workers for tickets, emergency alerts, and HVAC issues.
- Built a production **RAG service** using an **OpenSearch** vector store with document chunking and ingestion pipelines, exposed via a REST API for **semantic retrieval across 30+ compliance documents** (2,000+ pages), reducing **document review time from ~45 minutes to ~1 minute** and lowering **token consumption by 85%**.
- Streamlined **12 operational workflows** by building and deploying **n8n automations** that ingested **JIRA** data, predefined remediation tasks, and triggered Slack/email notifications, **auto-resolving 75% of incoming tickets**.

Ernst & Young

Mumbai, India

Software Engineer Intern

06/2023 - 10/2023

- Fine-tuned **Legal-BERT** on **200+ labeled regulations** for domain classification, achieving **87% accuracy**, and integrated a human-in-the-loop review system for low-confidence predictions, improving **overall classification accuracy to 94%**.
- Implemented an end-to-end service with **fine-grained access control** that integrated with eGazette to ingest regulatory updates, detect key differences using an **LLM**, and deliver classified updates to clients and internal teams, **reducing compliance processing time by 30%**.

PL Capital

Mumbai, India

Software Engineer, Quant Division

05/2021 - 03/2023

- Developed a Stock Grid tool comparing real-time and historical relative performance of **1,000+ stocks** against benchmarks to surface **buying opportunities during major market corrections**, improving backtested **strategy returns by 3.2%**.
- Built a **real-time stream processing pipeline** using **Apache Kafka** that ingested data from **50+ sources** and executed an in-house ML model to generate market sentiment scores with **end-to-end latency under 200 ms**.
- Redesigned and migrated a **high-traffic analytics dashboard** using **React, Highcharts, and Django**, reducing load times for large time ranges from **45 seconds to under 5 seconds** and enabling real-time strategy insights for **500+ analysts**.

PROJECTS

10 Days of Agents: Agentic AI Workflows, [GitHub](#)

- Engineered a modular agentic AI framework using **LangGraph**, integrating **PGVector/Chroma** for memory persistence, reusable tool registries, and OpenTelemetry tracing, delivering **10 end-to-end agent workflows in 10 days**.

PitchUp - Recruitment Platform, [GitHub](#)

- Created an AI-powered recruitment platform featuring video resumes and **Neo4j-based collaborative filtering**, improving **job matching by 12%** and reducing recruiter **review time by 30%** via **React** frontend deployed on **Google Cloud**.

Hemorrhoids Chatbot, [GitHub](#)

- Implemented a **medical RAG chatbot** with **Claude, FAISS, and red-flag detection**, evaluated against **Med42-8B** on safety and accuracy metrics.

Firestack - Developer Dashboard, [GitHub](#)

- Developed FireStack, a productivity dashboard featuring task management, job tracking, and **Google Calendar sync** using **React, Node.js, and PostgreSQL**, with **CI/CD automation** via **GitHub Actions** and **Docker** deployment.